Domingo Ranieri

City: Bologna, Italy ✻ *Date of birth:* 24/01/1998
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Profile

I hold a Master's degree in Theoretical Physics, where I developed a strong foundation in Python programming, numerical methods, and scientific data analysis using tools such as NumPy, Pandas, and Matplotlib.
This training strengthened my ability to model complex systems and solve analytical problems, skills I later applied in AI, data engineering, and distributed computing projects using frameworks such as TensorFlow, Keras, and PyTorch.
l My main focus is currently on LLM-based applications such as RAG pipelines, agents, and MCP integrations. I have additional experience in federated cloud orchestration and data lakehouse architectures, and have explored blockchain and quantum computing through personal projects.

Education

Master's Degree in Physics — *University of Bologna*
Theoretical Physics — *2020 - 2022*
Final grade: 110/110 cum laude
Thesis: Simulation of neuromuscular control using a quantum computer

Bachelor's Degree in Physics — *University of Bologna*
Physics — *2017 - 2020*
Final grade: 110/110 cum laude
Thesis: Fractal Universe model and Cosmic Acceleration

Work experience

ICSC — *Mar 2025 - Present*
Software Engineer — *Bologna, Italy*

- Research and development in AI, data engineering, and quantum computing
- Design and implementation of advanced software services
- Delivery of technical courses on AI, cloud, and data engineering
- Participation in conferences and collaborative research projects

ICSC-INFN — *Sep 2023 - Feb 2025*
Research Fellow — *Bologna, Italy*

- Development of AI and blockchain-based services
- Applied research in distributed and intelligent systems
- Participation in workshops and scientific events

Key Projects

Local Retrieval-Augmented Generation (RAG) System — *2026*
LLM / NLP Systems — *GitHub: study_rag*

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** system for question answering over custom document collections using **Milvus** as the vector database

- Designed a **context-aware chunking strategy** to improve semantic coherence and preserve document-level context during embedding and retrieval
- Implemented and compared **dense and hybrid retrieval pipelines**, achieving best performance with hybrid retrieval (MRR = 0.89, Recall@1 = 0.70, Recall@3 = 0.84, Recall@5 = 0.89)
- Developed an **LLM-as-a-judge evaluation framework** to assess generated answer quality beyond retrieval metrics, focusing on correctness, relevance, and completeness
- Engineered a fully **local inference pipeline using Ollama and Hugging Face models**, enabling end-to-end execution without external API dependencies
- Designed the system to be **modular and API-agnostic**, allowing seamless integration with external providers such as OpenAI or Anthropic

Secure MCP Chatbot Host Platform *2026*
Backend / LLM Systems / Security Engineering

- Developed a **FastAPI-based chatbot hosting platform** enabling interaction with both **public and private MCP (Model Context Protocol) servers**
- Followed and implemented key security principles from the **OWASP MCP Security Cheat Sheet**,including controlled tool exposure, role based access restriction, and secure request handling
- Integrated **Keycloak (OAuth2)** for authentication and authorization, with role-based access management
- Utilized **PostgreSQL** for persistent data storage and **Redis** for caching and session optimization
- Built a lightweight frontend using **HTML, CSS, and JavaScript** for chat interaction and tool execution visualization
- Developed the system using a fully **Dockerized architecture**, designed to be Kubernetes-ready for scalable deployment
- Integrated a locally hosted **Ollama LLM runtime** to enable private inference without external API dependencies

Technical skills

Programming	Python, Go, Bash, C++
ML/AI	Scikit-learn, TensorFlow, PyTorch, Keras
Data Engineering	Kafka, Airflow, MinIO
Cloud & DevOps	Docker, Kubernetes, Ansible
Databases	PostgreSQL, MongoDB, MySQL, Redis
Web	Flask, FastAPI, HTML, CSS, JavaScript, Streamlit
Tools	Claude Code, Codex, MLflow, Grafana, Prometheus, Loki, Git

Languages

Italian	Native
English	C1

Communication and interpersonal skills

Effective presentation and public speaking

Strong technical writing and documentation skills

Collaborative teamwork in multidisciplinary environments

Ability to explain complex concepts clearly